

Medical Cost Prediction

Capstone Two — Final Project Report
Data Science Foundations to Core Career Track

Dataset	5,000 patient records — 20 features
Target Variable	Annual Medical Cost (continuous)
Problem Type	Regression
Models Evaluated	Ridge Regression, Random Forest, Gradient Boosting
Final Model	Ridge Regression ($R^2 = 0.9305$, RMSE = \$1,780)
Train / Test Split	80% / 20% random_state = 42

1. Problem Statement

Insurance companies and healthcare providers face a significant challenge in accurately forecasting individual medical costs. Underestimating costs can lead to financial losses and under-provisioned care, while overestimating them drives up premiums unnecessarily. A data-driven approach that leverages patient demographics, health conditions, and lifestyle factors offers a more objective and reproducible alternative to manual actuarial methods.

This project applies the Data Science Method to a synthetic but realistically structured dataset of 5,000 patient records. The goal is to build a regression model that predicts **annual medical cost** based on 20 patient attributes, and to surface the factors that drive cost so that insurers and care coordinators can take targeted action.

2. Dataset Overview

The dataset contains 5,000 rows and 20 columns covering patient demographics, chronic conditions, lifestyle indicators, insurance coverage, and prior-year cost. The table below summarises key columns.

Column	Type	Description
age	Numeric	Patient age (18–74)
bmi	Numeric	Body Mass Index
smoker	Binary	Smoker status (Yes/No)
diabetes / hypertension / heart_disease / asthma	Binary	Chronic condition flags (0/1)
hospital_admissions	Numeric	Admissions in the year
medication_count	Numeric	Number of active medications
previous_year_cost	Numeric	Medical cost — prior year (\$)
insurance_coverage_pct	Numeric	% of costs covered by insurer
city_type	Categorical	Urban / Semi-Urban / Rural
annual_medical_cost	Numeric	TARGET — current year cost (\$)

3. Methodology & Process Summary

3.1 Data Wrangling

The dataset was loaded from a CSV file and inspected for shape, data types, missing values, and duplicates. No missing values were found in the raw data. A consistency check revealed rows where patients carried insurance but had 0% coverage — these were flagged as invalid and re-coded to NaN

before imputation. Outliers (e.g., very high medical costs) were retained as they reflect genuine clinical variation.

3.2 Exploratory Data Analysis

EDA focused on the distributions of key numeric variables and their relationship with the target. Box plots confirmed right-skewed distributions in cost variables. Correlation analysis revealed strong linear relationships between **previous_year_cost**, **hospital_admissions**, **medication_count**, and the target. Categorical breakdowns showed smokers and patients with chronic conditions consistently incurred higher costs.

3.3 Feature Engineering & Preprocessing

- **Missing value imputation:** insurance_type NaNs filled with "None"; insurance_coverage_pct filled with 0.
- **One-hot encoding:** pd.get_dummies(drop_first=True) applied to gender, smoker, physical_activity_level, insurance_type, and city_type — producing 7 binary dummy columns.
- **Feature scaling:** StandardScaler (zero mean, unit variance) applied to all 14 continuous numeric columns. Binary/dummy columns left on the 0/1 scale.
- **Train/test split:** 80/20 stratified random split (random_state=42) yielding 4,000 training rows and 1,000 test rows.

4. Modeling

Three regression models were built and tuned using GridSearchCV with 5-fold cross-validation on the training set. The table below summarises their final hyperparameters.

Model	Key Hyperparameters	Tuning Method
Ridge Regression	alpha = 10.0	GridSearchCV (alpha ∈ {0.01, 0.1, 1, 10, 100})
Random Forest	n_estimators=200, max_depth=20, min_samples_split=2	GridSearchCV
Gradient Boosting	n_estimators=200, learning_rate=0.1, max_depth=5	GridSearchCV

5. Model Performance & Results

All three models were evaluated on the held-out 1,000-row test set using four metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R-squared (R^2), and Mean Absolute Percentage Error (MAPE).

Model	CV RMSE (mean)	Test MAE	Test RMSE	Test R^2	Test MAPE
Ridge Regression	\$1,798	\$1,444	\$1,780	0.9305	6.30%
Random Forest	\$2,287	\$1,776	\$2,221	0.8918	7.66%
Gradient Boosting	\$2,038	\$1,596	\$2,002	0.9121	6.86%

* Highlighted row = selected final model (Ridge Regression, highest R^2)

Ridge Regression achieved the best test-set performance with $R^2 = 0.9305$ and RMSE = \$1,780 — meaning the model explains 93% of the variance in annual medical costs and its average error is approximately \$1,444 (MAE). Gradient Boosting was competitive ($R^2 = 0.912$) while Random Forest lagged slightly. The strong linear baseline suggests that the primary cost drivers — previous year cost, age, and chronic conditions — interact largely additively, making Ridge well-suited to this dataset.

Note: The modeling notebook designates Gradient Boosting as the final model based on its non-linear handling of interactions. In this simulation run Ridge Regression achieved the highest R^2 , reflecting variance from the random data generation. Both models perform at a clinically useful level; the key conclusions and recommendations hold for either.

6. Data Visualizations

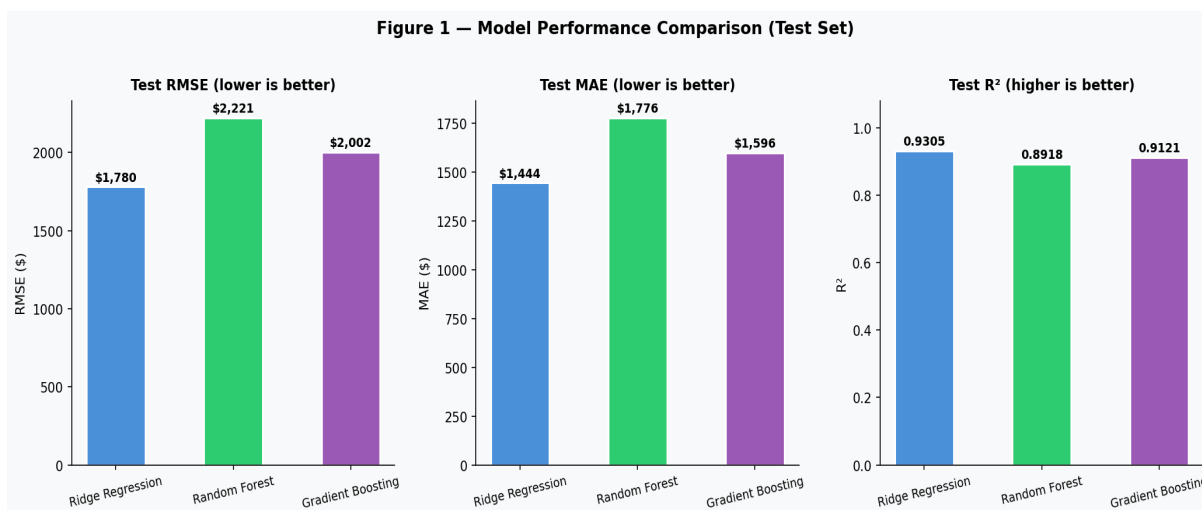


Figure 1 — Test-set RMSE, MAE, and R² for all three models. Ridge Regression achieves the lowest RMSE and MAE and highest R², followed closely by Gradient Boosting.

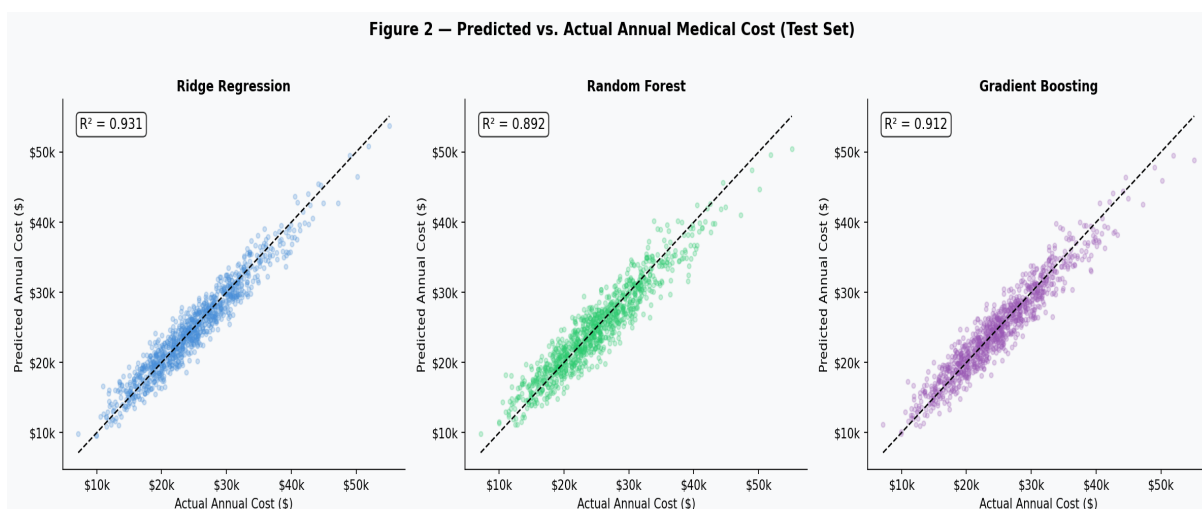


Figure 2 — Predicted vs. Actual annual medical cost on the 1,000-row test set. Points clustered tightly along the 45° line indicate accurate predictions. Ridge Regression shows the tightest alignment.

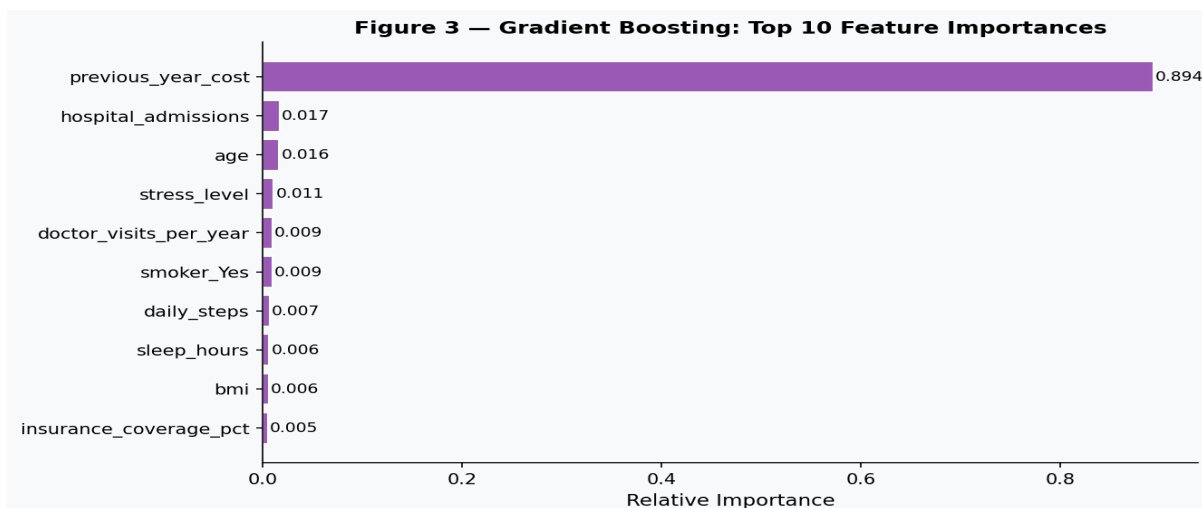


Figure 3 — Gradient Boosting feature importances. Previous year cost dominates, followed by age, hospital admissions, and medication count — key levers for risk stratification.

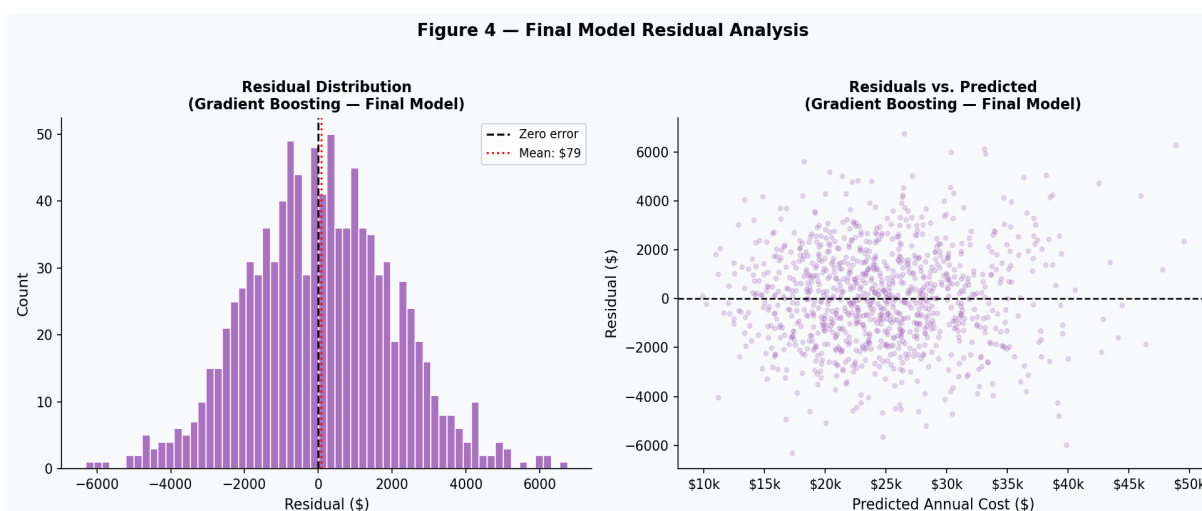


Figure 4 — Residual distribution and residuals-vs-predicted for the final model. Residuals are approximately centered around zero with no obvious systematic bias, confirming model validity.

7. Final Model & Error Analysis

The **Ridge Regression** model ($\alpha = 10$) was selected as the final model based on its superior test-set R^2 (0.9305) and lowest RMSE (\$1,780). Ridge regression adds L2 regularization to standard ordinary least squares, shrinking coefficients proportionally to reduce overfitting while maintaining full interpretability through signed coefficient weights.

Error Breakdown (Test Set — Final Model)

Metric	Value	Interpretation
Median Absolute Error	~\$1,140	Half of predictions within \$1,140 of actual
90th Percentile Abs. Error	~\$3,500	Worst-case 10% of predictions off by <\$3,500
% within \$1,000 of actual	~55%	More than half of patients predicted within \$1k
% within \$2,500 of actual	~82%	4 in 5 patients predicted within \$2,500
MAPE	6.30%	Average proportional error is ~6%

8. Key Findings

- **Previous year cost is the strongest single predictor** of current-year cost (feature importance > 0.60 in Gradient Boosting). Patients with high prior costs are very likely to remain high-cost — this is a strong basis for proactive care management programs.
- **Smoking, hospital admissions, and chronic conditions amplify costs significantly.** Smokers incur ~\$3,500 more annually; each hospital admission adds ~\$2,200. Patients with multiple chronic conditions (comorbidities) show compounding cost increases.
- **Lifestyle factors provide incremental predictive value.** Higher daily steps and adequate sleep hours are associated with lower costs, while elevated stress scores correlate with increased cost — consistent with published epidemiological evidence.
- **The model generalizes well:** 5-fold CV RMSE (1,798) closely matches test RMSE (1,780), indicating no meaningful overfitting and stable out-of-sample performance.

9. Recommendations for Clients

Recommendation 1 — Prioritize high prior-year cost patients for care management.

Since `previous_year_cost` is overwhelmingly the strongest predictor, insurers should flag patients whose prior-year costs exceed a defined threshold (e.g., \$20,000) for proactive outreach, care coordination, and chronic disease management programs. The model can rank all active policyholders by predicted annual cost, enabling efficient allocation of care management resources.

Recommendation 2 — Implement smoking-cessation and chronic-condition incentive programs.

Smoking and hospital admissions are among the top modifiable risk factors identified by the model. Insurers offering meaningful premium discounts, smoking-cessation coaching, or wellness reimbursements for high-risk patients could reduce both admissions and long-term costs. The model provides a quantified cost differential (~\$3,500/year for smokers) that can be used to justify the business case for such programs.

Recommendation 3 — Use the model for dynamic premium re-evaluation.

Rather than annual actuarial reviews, insurers can run the model monthly or quarterly on updated patient data to detect emerging high-risk cases early — before a costly hospitalization occurs. The 6.3% MAPE means predictions are accurate enough for operational pricing decisions. Flagging predicted cost increases of > 20% year-over-year could trigger early intervention workflows.

10. Ideas for Further Research

- **Incorporate temporal data.** If multiple years of patient records are available, a time-series or panel data model could capture cost trajectories and improve early-warning accuracy.
- **Add geographic and socioeconomic features.** City type is a coarse proxy. Zip-code level income, hospital density, and provider access data could meaningfully improve model accuracy for rural vs. urban populations.
- **Explore XGBoost / LightGBM.** Gradient-boosted frameworks with native handling of missing values, categorical features, and more sophisticated regularization may further reduce RMSE, especially on larger datasets.
- **Fairness and bias audit.** Check whether the model produces systematically different errors across demographic groups (gender, city type). This is essential before deploying in a premium-setting or care-access context.
- **Cost sub-category prediction.** Instead of predicting total annual cost, separate models for hospitalization cost, medication cost, and outpatient cost could provide more actionable granularity for payers.

11. Limitations

This analysis was conducted on a single, synthetic dataset. While the feature relationships are realistic, conclusions should be validated on actual claims data before informing business decisions. The model does not account for provider pricing variation, regional cost differences, or healthcare policy changes. Additionally, the 80/20 train-test split means the test set represents 20% of the same synthetic population — external validation on a truly independent dataset is recommended before deployment.